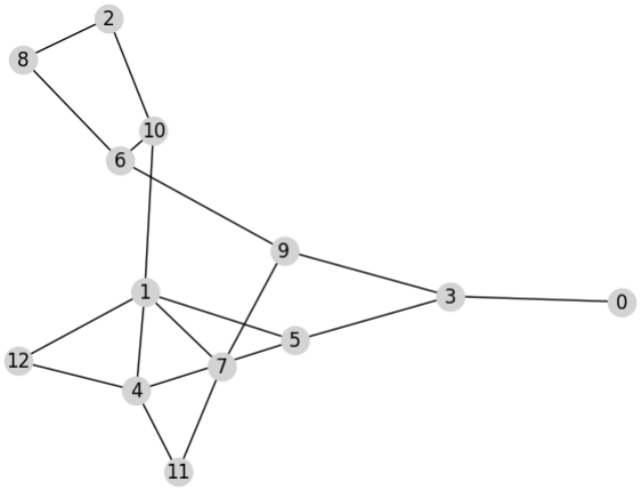
Indian Institute of Technology Bhilai
Department of Computer Science Engineering
Network Science (CSL601/CS552)
Date: April 27, 2024

Timing: 2:30 PM to 5:30 PM Max mark: 100

1. (a) Identify the weak communities in the given network. Can any of these weak communities be considered a strong community? If yes, what are the conditions for it? **(10)**

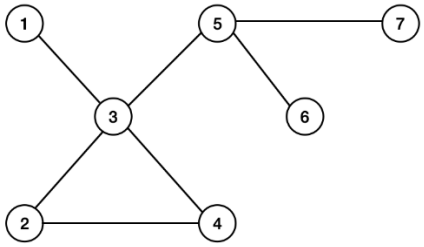


(b) Explain different types of cluster similarity metrics. Give examples. **(10)**

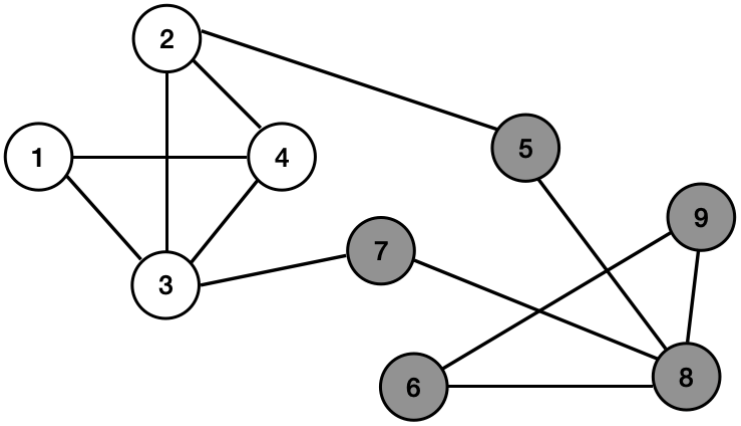
2. Briefly explain.
   (a) Triadic closure **(2)**
   (b) Reasons for triadic closure **(4)**
   (c) How does triadic closure play a role in the formation of social networks? **(4)**

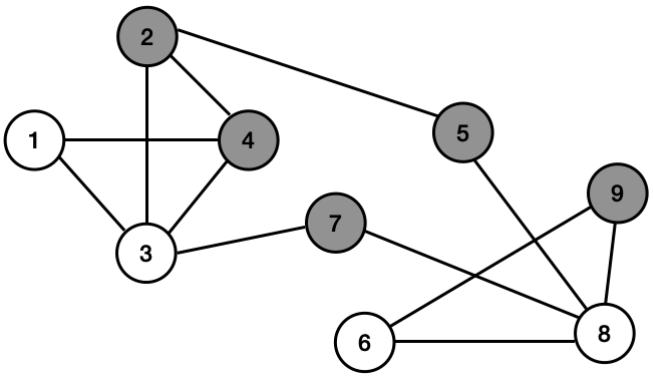
3. Isha and Disha are two friends in college, they have 5 out of 6 classes in common, they study together, shop together, etc so they can be considered close friends. When Isha was in school, she was in a class of 50 students out of which Ankita, Dev, and Sushant were her close friends and she still hangs out with them. When Disha was in school, she was close friends with Rohit and Riya but they are not in contact anymore. Is Disha and Isha's friendship a local bridge if their social circles are limited to their school friends? Explain. **(5)**

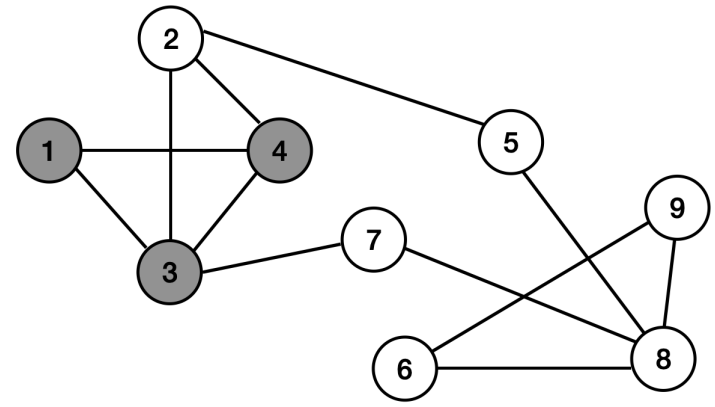
4. Perform divisive clustering on the following network and draw the resulting dendrogram. **(40)**

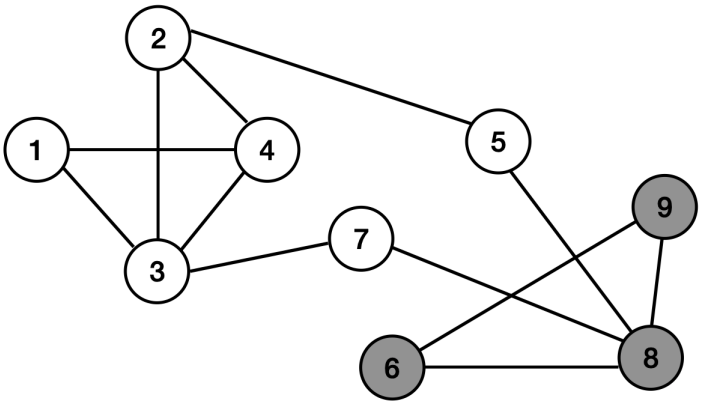


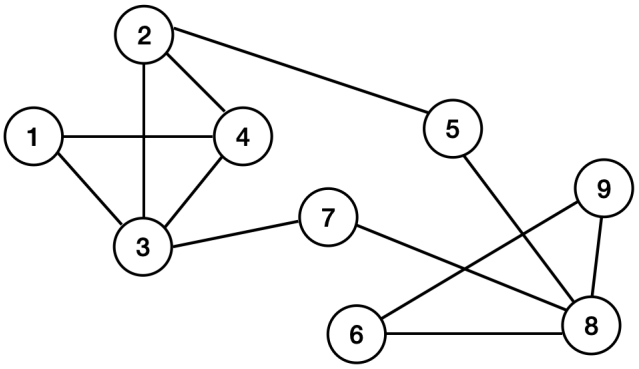
5. Compute modularity for each of the below given partitions. Which partition has the highest modularity? **(20)**











6. Discuss the three classic epidemic models - SI, SIS, and SIR. How do these models describe the dynamics of infectious disease spread within a population, and what is the significance of the parameters in their differential equations? (You must write down the coupled differential equations for these models and explain it.) **(15)**